Exercise 6.8 SOLUTION

1. Find the 8th term of the arithmetico-geometric sequence, where the arithmetic part is $1,4,7,\ldots$ and the geometric part is $5,10,20,\ldots$

Sol: Identify the arithmetic part and its general term:
The arithmetic sequence is $1,4,7,\ldots$
The common difference is $d=4-1=3.$
The nth term of an arithmetic sequence is $a_n=a_1+(n-1)d$
Here, $a_1=1$ & $d=3$, so $a_n=1+(n-1)3=1+3n-3=3n-2$

i) **Identify the geometric part and its general term:**
The geometric sequence is $5,10,20,\ldots$
The common ratio is $r=\frac{10}{5}=2$
The nth term of a geometric sequence is $g_n=g_1.r^{n-1}$
Here, $g_1=5$ & $r=2$ so, $g_n=5.2^{n-1}$

ii) From the general term of the arithmetico-geometric sequence:
The nth term of the arithmetico-geometric sequence, T_n is the product of the nth terms of the arithmetic and geometric parts.
$T_n=a_n.g_n=(3n-2).(5.2^{n-1})$

iii) **Calculate the 8th term (T_8):**
Substitute $n=8$ into the formula for T_n
$T_8=(3(8)-2).(5.2^{8-1})=(24-2).(5.2^7)$
$T_8=(22).(5).(128)=14080$
Therefore, the 8th term of the arithmetico-geometric sequence is 14080.

2. Find the nth term of the arithmetic-geometric sequence, where the arithmetic part is $3,7,11\ldots$ and the geometric part is $2,6,18,\ldots.$

Sol:

i) **Analyze the arithmetic part:**
The arithmetic sequence is $3,7,11,\ldots$
The first term is $a_1=3$
The common difference is $d=7-3=4.$
The nth term of an arithmetic sequence is given by the formula:
$a_n=a_1+(n-1)d$ substituting the values of a_1 and d:
$a_n=3+(n-1)4=3+4n-4=4n-1$

ii) **Analyze the geometric part:**
The geometric sequence is $2,6,18,\ldots$
The first term is $g_1=2.$
The common ratio is $r=\frac{6}{2}=3.$
The nth term of a geometric sequence is given by the formula:
$g_n=g_1.r^{n-1}$ Substituting the values of g_1 and r
$g_n=2.3^{n-1}$

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iii) **Combine the arithmetic and geometric parts:**

The nth term of the arithemetic0-geometric sequence, T_n, is the product of the nth terms of the arithmetic and geometric parts:

$T_n = a_n.g_n$ substituting the expressions, we found for a_n and g_n:

$T_n = (4n-1).(2.3^{n-1})$

$T_n = 2(4n-1).3^{n-1}$

Thus, the nth term of the arithmetico-geometric sequence is $2(4n-1).3^{n-1}$.

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3. Consider the arithmetic-geometric sequence defined by arithmetic part: $a_{n+1} = 2n+5$ and geometric part: $b_{n-2} = \frac{1}{9}(-3)^n$. Find the nth term and the sum of first three terms of the arithmetic-geometric sequence.

Sol:

i) **Arithmetic part:**

Given $a_{n+1} = 2n+5$

Substitute $n-1$ for b:

$a_{(n-1)+1} = 2(n-1)+5$

$a_n = 2n-2+5$

$a_n = 2n+3$

First three terms of the arithmetic part:

$a_1 = 2(1)+3 = 5$

$a_2 = 2(2)+3 = 7$

$a_3 = 2(3)+3 = 9$

ii) **Geometric part:**

Given $b_{n-2} = \frac{1}{9}(-3)^n$

Replace n by n+2

$b_{n+2-2} = \frac{1}{9}(-3)^{n+2} \Rightarrow b_n = \frac{1}{9}(-3)^{n+2}$

$b_1 = \frac{1}{9}(-3)^{1+2} = \frac{1}{9}(-3)^3 = \frac{1}{9}(-27) = -3$

$b_2 = \frac{1}{9}(-3)^{2+2} = \frac{1}{9}(-3)^4 = \frac{1}{9}(81) = 9$

$b_3 = \frac{1}{9}(-3)^{3+2} = \frac{1}{9}(-3)^5 = \frac{1}{9}(-243) = -27$

iii) **Nth term of the arithmetico-geometric sequence (T_n)**

$$T_n = a_n.b_n = (2n+3)\left(\frac{1}{9}(-3)^{n+2}\right) = (2n+3)\left[\frac{1}{9}(-3)^2.(-3)^n\right] = (2n+3)\left[\frac{1}{\cancel{9}}\cancel{(9)}.(-3)^n\right] = (2n+3)(-3)^n$$

iv) **First three terms of the arithmetico-geometric sequence:**

$T_1 = a_1.b_1 = (5)(-3) = -15$

$T_2 = a_2.b_2 = (7)(9) = 63$

$T_3 = a_3.b_3 = (9)(-27) = -243$

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v) **Sum of the first three terms (S_3)**

$S_3 = T_1 + T_2 + T_3 = -15 + 63 - 243 = -195$

The final answer is $T_n = (2n+3)(-3)^n, S_3 = -195$

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4. **Sum to n terms the following series:**

Sol:

i) $1.2 + 3.4 + 5.8 + 7.16 + \ldots$

Step 1: Identify the kth term

- The first factors $(1,3,5,7,\ldots)$ from an arithmetic progression with the kth term arithmetic progression with the kth term $1+(k-1)2 = 2k-1$.
- The second factors $(2,4,8,16,\ldots)$ from an geometric progression with the kth term progression with the kth term $2.2^{k-1} = 2^k$.

  The kth term, $T_k = (2k-1).2^k$.

Step 2: Write sum of the first n terms

Let S_n be the sum of the first n terms:

$S_n = 1.2^1 + 3.2^2 + 5.2^3 + \ldots + (2n-1)2^n$

Step 3: Multiply the sum by the common ratio of the geometric part (which is 2)

$2S_n = 1.2^2 + 3.2^3 + 5.2^4 + \ldots + (2n-3)2^n + (2n-1)2^{n+1}$

Step 4: Subtract the original sum (S_n) from the multiplied sum $(2S_n)$

$$(2S_n - S_n) = \left[(2n-1)2^{n+1}\right] - \left[1.2^1 + (3-1)2^2 + (5-3)2^3 + \ldots + \left((2n-1)-(2n-3)\right)2^n\right]$$

$$S_n = (2n-1)2^{n+1} - \left[2 + 2.2^2 + 2.2^3 + \ldots + 2.2^n\right]$$

Step 5: Simplify the resulting series

The series in the brackets is a constant term plus a geometric series:

$2 + 2.2^2 + 2.2^3 + \ldots + 2.2^n = 2 + 2\left(2^2 + 2^3 + \ldots + 2^n\right)$

Step 6: Find the sum of the geometric series:

The geometric series $2^2 + 2^3 + \ldots + 2^n$ has the first term $a = 2^2 = 4$, common ratio $r = 2$, and $n-1$ terms.

The sum is: $\dfrac{a\left(r^{n-1}-1\right)}{r-1} = \dfrac{4\left(2^{n-1}-1\right)}{2-1} = 4\left(2^{n-1}-1\right) = 2^{n+1} - 4$

Step 7: Substitute the sum of the geometric series back into the equation for S_n

$S_n = (2n-1)2^{n+1} - \left[2 + 2\left(2^{n+1} - 4\right)\right]$

$S_n = (2n-1)2^{n+1} - \left[2 + 2^{n+2} - 8\right]$

$S_n = (2n-1)2^{n+1} - \left[2^{n+2} - 6\right]$

Step 8: Simplify the final expression

$S_n = 2n.2^{n+1} - 1.2^{n+1} - 2^{n+2} + 6$

$S_n = n.2^{n+2} - 2^{n+1} - 2.2^{n+1} + 6$

$S_n = n.2^{n+2} - 3.2^{n+1} + 6$

$S_n = 2^{n+1}(2n-3) + 6 = 6 + (4n-6)2^n$

Therefore, the sum to n terms of the series is $= 6 + (4n-6)2^n$

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ii) $2.3+4.3^2+6.3^3+8.3^4+\ldots$

The kth term of this series is $T_k=(2k).3^k$.

Let $S_n=2.3^1+4.3^2+6.3^3+\ldots+2n.3^n$

Multiply S_n by 3:

$3S_n=2.3^2+4.3^3+6.3^4+\ldots+2(n-1).3^n+2n.3^{n+1}$

Subtract S_n from $3S_n$:

$3S_n-S_n=2n.3^{n+1}-\left(2.3^1+(4-2).3^2+(6-4).3^3+\ldots+(2n-2(n-1)).3^n\right)$

$2S_n=2n.3^{n+1}-\left(6+2.3^2+2.3^3+\ldots+2.3^n\right)$

$2S_n=2n.3^{n+1}-\left(6+2\left(3^2+3^3+\ldots+3^n\right)\right)$

The sum of the geometric series $3^2+3^3+\ldots+3^n=\dfrac{3^2\left(3^{n-1}-1\right)}{3-1}=\dfrac{9\left(3^{n-1}-1\right)}{2}=\dfrac{3^{n+1}-9}{2}$

Substitute this back into the expression for $2S_n$:

$2S_n=2n.3^{n+1}-\left(6+2.\dfrac{3^{n+1}-9}{2}\right)$

$2S_n=2n.3^{n+1}-\left(6+3^{n+1}-9\right)$

$2S_n=2n.3^{n+1}-3^{n+1}+3$

$S_n=\dfrac{(2n-1)3^{n+1}+3}{2}$

$\dfrac{3}{2}\left[(2n-1)3^n+1\right]=\dfrac{3}{2}\left[2n.3^n-3^n+1\right]$

$\dfrac{3}{2}\left[2n.3^n-3^n+1+3n.3^n-3n.3^n\right]\Rightarrow\dfrac{3}{2}\left[1-3^n-n.3^n+n.3^{n+1}\right]$

$\dfrac{3}{2}\left[1-(n+1)3^n+n.3^{n+1}\right]$

iii) $2+\dfrac{5}{4}+\dfrac{8}{4^2}+\dfrac{11}{4^3}+\ldots$

The kth term of this series is $T_k=\dfrac{2+(k-1)3}{4^{k-1}}=\dfrac{3k-1}{4^{k-1}}$

Let $S_n=2+\dfrac{5}{4}+\dfrac{8}{4^2}+\ldots+\dfrac{3n-1}{4^{n-1}}$.

Multiply S_n by $\dfrac{1}{4}$:

$\dfrac{1}{4}S_n=\dfrac{2}{4}+\dfrac{5}{4^2}+\dfrac{8}{4^3}+\ldots+\dfrac{3n-4}{4^{n-1}}+\dfrac{3n-1}{4^n}$.

Subtract $\dfrac{1}{4}S_n$ from S_n:

$S_n-\dfrac{1}{4}S_n=2+\left(\dfrac{5}{4}-\dfrac{2}{4}\right)+\left(\dfrac{8}{4^2}-\dfrac{5}{4^2}\right)+\ldots+\left(\dfrac{3n-1}{4^{n-1}}-\dfrac{3n-4}{4^{n-1}}\right)-\dfrac{3n-1}{4^n}$

$\dfrac{3}{4}S_n=2+\dfrac{3}{4}+\dfrac{3}{4^2}+\ldots+\dfrac{3}{4^{n-1}}-\dfrac{3n-1}{4^n}\Rightarrow\dfrac{3}{4}S_n=2+3\left(\dfrac{1}{4}+\dfrac{1}{4^2}+\ldots+\dfrac{1}{4^{n-1}}\right)-\dfrac{3n-1}{4^n}$

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The sum of the geometric series $\frac{1}{4}+\frac{1}{4^2}+\ldots+\frac{1}{4^{n-1}}=\frac{\frac{1}{4}\left(1-\left(\frac{1}{4}\right)^{n-1}\right)}{1-\frac{1}{4}}=\frac{\frac{1}{4}\left(1-\frac{1}{4^{n-1}}\right)}{\frac{3}{4}}=\frac{1}{3}\left(1-\frac{1}{4^{n-1}}\right)$

Substitute this back into the expression for $\frac{3}{4}S_n$:

$$\frac{3}{4}S_n=2+3.\frac{1}{3}\left(1-\frac{1}{4^{n-1}}\right)-\frac{3n-1}{4^n}$$

$$\frac{3}{4}S_n=2+1-\frac{1}{4^{n-1}}-\frac{3n-1}{4^n}$$

$$\frac{3}{4}S_n=3-\frac{4}{4^n}-\frac{3n-1}{4^n}$$

$$\frac{3}{4}S_n=3-\frac{3n+3}{4^n}$$

$$S_n=\frac{4}{3}\left(3-\frac{3(n+1)}{4^n}\right)$$

$$S_n=4-\frac{4(n+1)}{4^n}=4-\frac{n+1}{4^{n-1}}=4-\frac{4(1+n)}{4^n}=4-\frac{12(1+n)}{3.4^n}=4-\frac{12+12n}{3.4^n}=4-\frac{16+(12n-4)}{3.4^n}$$

$$=4-\frac{16}{3.4^n}-\frac{12n-4}{3.4^n}=4-\frac{4}{3}.\frac{4}{4^n}-\frac{4(3n-1)}{3.4^n}=4-\frac{4}{3}\left(\frac{1}{4}\right)^{n-1}-\frac{4}{3}(3n-1)\left(\frac{1}{4}\right)^n$$

iv) $1+\frac{3}{5}+\frac{5}{5^2}+\frac{7}{5^3}+\ldots$

The kth term of this series is $T_k=\frac{1+(k-1)2}{5^{k-1}}=\frac{2k-1}{5^{k-1}}$

Let $S_n=1+\frac{3}{5}+\frac{5}{5^2}+\ldots+\frac{2n-1}{5^{n-1}}$.

Multiply S_n by $\frac{1}{5}$:

$$\frac{1}{5}S_n=\frac{1}{5}+\frac{3}{5^2}+\frac{5}{5^3}+\ldots+\frac{2n-3}{5^{n-1}}+\frac{2n-1}{5^n}.$$

Subtract $\frac{1}{5}S_n$ from S_n:

$$S_n-\frac{1}{5}S_n=1+\left(\frac{3}{5}-\frac{1}{5}\right)+\left(\frac{5}{5^2}-\frac{3}{5^2}\right)+\ldots+\left(\frac{2n-1}{5^{n-1}}-\frac{2n-3}{5^{n-1}}\right)-\frac{2n-1}{5^n}$$

$$\frac{4}{5}S_n=1+\frac{2}{5}+\frac{2}{5^2}+\ldots+\frac{2}{5^{n-1}}-\frac{2n-1}{5^n}$$

$$\frac{4}{5}S_n=1+2\left(\frac{1}{5}+\frac{1}{5^2}+\ldots+\frac{1}{5^{n-1}}\right)-\frac{2n-1}{5^n}$$

The sum of the geometric series $\frac{1}{5}+\frac{1}{5^2}+\ldots+\frac{1}{5^{n-1}}=\frac{\frac{1}{5}\left(1-\left(\frac{1}{5}\right)^{n-1}\right)}{1-\frac{1}{5}}=\frac{\frac{1}{5}\left(1-\frac{1}{5^{n-1}}\right)}{\frac{4}{5}}=\frac{1}{4}\left(1-\frac{1}{5^{n-1}}\right)$

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Substitute this back into the expression for $\frac{4}{5}S_n$:

$$\frac{4}{5}S_n = 1 + 2.\frac{1}{4}\left(1 - \frac{1}{5^{n-1}}\right) - \frac{2n-1}{5^n}$$

$$\frac{4}{5}S_n = 1 + \frac{1}{2} - \frac{1}{2.5^{n-1}} - \frac{2n-1}{5^n}$$

$$\frac{4}{5}S_n = \frac{3}{2} - \frac{5.}{2.5^n} - \frac{2(2n-1)}{2.5^n}$$

$$\frac{4}{5}S_n = \frac{3}{2} - \frac{5+4n-2}{2.5^n} = \frac{3}{2} - \frac{4n+3}{2.5^n}$$

$$S_n = \frac{5}{4}\left(\frac{3}{2} - \frac{4n+3}{2.5^n}\right) = \frac{15}{8} - \frac{5(4n+3)}{8.5^n} = \frac{15}{8} - \frac{4n+3}{8.5^{n-1}}$$

$$= \frac{15}{8} - \frac{4n+3}{8}\left(\frac{1}{5}\right)^{n-1} = \frac{15}{8} - \left[\frac{2n-1}{4} + \frac{5}{8}\right]\left(\frac{1}{5}\right)^{n-1} = \frac{15}{8} - \left(\frac{2n-1}{4}\right)\left(\frac{1}{5}\right)^{n-1} - \frac{5}{8}\left(\frac{1}{5}\right)^{n-1}$$

$$= \frac{15}{8} - \left(\frac{2n-1}{4}\right).5\left(\frac{1}{5}\right)^{n} - \frac{5}{8}.\left(\frac{1}{5}\right)^{n-1} = \frac{15}{8} - \frac{5(2n-1)}{4}\left(\frac{1}{5}\right)^{n} - \frac{5}{8}\left(\frac{1}{5}\right)^{n-1}$$

v) $\mathbf{1 + \frac{4}{3} + \frac{7}{9} + \frac{10}{27} + \ldots}$

The kth term of this series is $T_k = \frac{1+(k-1)3}{3^{k-1}} = \frac{3k-2}{3^{k-1}}$

Let $S_n = 1 + \frac{4}{3} + \frac{7}{3^2} + \ldots + \frac{3n-2}{3^{n-1}}$.

Multiply S_n by $\frac{1}{3}$:

$$\frac{1}{3}S_n = \frac{1}{3} + \frac{4}{3^2} + \frac{7}{3^3} + \ldots + \frac{3n-5}{3^{n-1}} + \frac{3n-2}{3^n}.$$

Subtract $\frac{1}{3}S_n$ from S_n:

$$S_n - \frac{1}{3}S_n = 1 + \left(\frac{4}{3} - \frac{1}{3}\right) + \left(\frac{7}{3^2} - \frac{4}{3^2}\right) + \ldots + \left(\frac{3n-2}{3^{n-1}} - \frac{3n-5}{3^{n-1}}\right) - \frac{3n-2}{3^n}$$

$$\frac{2}{3}S_n = 1 + \frac{3}{3} + \frac{3}{3^2} + \ldots + \frac{3}{3^{n-1}} - \frac{3n-2}{3^n} \Rightarrow \frac{2}{3}S_n = 1 + 3\left(\frac{1}{3} + \frac{1}{3^2} + \ldots + \frac{1}{3^{n-1}}\right) - \frac{3n-2}{3^n}$$

The sum of the geometric series $\frac{1}{3} + \frac{1}{3^2} + \ldots + \frac{1}{3^{n-1}} = \frac{\frac{1}{3}\left(1 - \left(\frac{1}{3}\right)^{n-1}\right)}{1 - \frac{1}{3}} = \frac{\frac{1}{3}\left(1 - \frac{1}{3^{n-1}}\right)}{\frac{2}{3}} = \frac{1}{2}\left(1 - \frac{1}{3^{n-1}}\right)$

Substitute this back into the expression for $\frac{2}{3}S_n$:

$$\frac{2}{3}S_n = 1 + 3.\frac{1}{2}\left(1 - \frac{1}{3^{n-1}}\right) - \frac{3n-2}{3^n} \Rightarrow \frac{2}{3}S_n = 1 + \frac{3}{2} - \frac{3}{2.3^{n-1}} - \frac{3n-2}{3^n}$$

$$\frac{2}{3}S_n = \frac{5}{2} - \frac{9}{2.3^n} - \frac{2(3n-2)}{2.3^n}.$$

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$$\frac{2}{3}S_n = \frac{5}{2} - \frac{9+6n-4}{2.3^n} = \frac{5}{2} - \frac{6n+5}{2.3^n}$$

$$S_n = \frac{3}{2}\left(\frac{5}{2} - \frac{6n+5}{2.3^n}\right) = \frac{15}{4} - \frac{3(6n+5)}{4.3^n} = \frac{15}{4} - \frac{6n+5}{4.3^{n-1}}$$

$$= \frac{15}{4} - \frac{6n+5}{4}\left(\frac{1}{3}\right)^{n-1} = \frac{15}{4} - \left[\frac{3n-2}{2} + \frac{9}{4}\right]\left(\frac{1}{3}\right)^{n-1} = \frac{15}{4} - \frac{3n-2}{2}.\left(\frac{1}{3}\right)^{n} - \frac{9}{4}\left(\frac{1}{3}\right)^{n-1}$$

$$= \frac{15}{4} - \frac{3n-2}{2}.3.\left(\frac{1}{3}\right)^{n} - \frac{9}{4}\left(\frac{1}{3}\right)^{n-1} = \frac{15}{4} - \frac{3}{2}(3n-2)\left(\frac{1}{3}\right)^{n} - \frac{9}{4}\left(\frac{1}{3}\right)^{n-1}$$

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5. **Sum the following infinite series:**

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i) $\mathbf{1+\frac{3}{2}+\frac{5}{4}+\frac{7}{8}+\ldots}$

Sol: Let $S = 1+\frac{3}{2}+\frac{5}{4}+\frac{7}{8}+\ldots$

Multiply both sides by $\frac{1}{2}$:

$$\frac{1}{2}S = \frac{1}{2}+\frac{3}{4}+\frac{5}{8}+\frac{7}{16}+\ldots$$

Now, subtract the second equation from the first:

$$S - \frac{1}{2}S = \left(1+\frac{3}{2}+\frac{5}{4}+\frac{7}{8}+\ldots\right) - \left(\frac{1}{2}+\frac{3}{4}+\frac{5}{8}+\frac{7}{16}+\ldots\right)$$

$$\frac{1}{2}S = 1+\left(\frac{3}{2}-\frac{1}{2}\right)+\left(\frac{5}{4}-\frac{3}{4}\right)+\left(\frac{7}{8}-\frac{5}{8}\right)+\ldots$$

$$\frac{1}{2}S = 1+\frac{2}{2}+\frac{2}{4}+\frac{2}{8}+\ldots$$

$$\frac{1}{2}S = 1+1+\frac{1}{2}+\frac{1}{4}+\ldots \text{ ---------(i)}$$

The terms $\frac{1}{2}+\frac{1}{4}+\ldots$ from an infinite geometric series with the first term $a=\frac{1}{2}$ and the common ratio $r=\frac{1}{2}$.

The sum of such a series is $\frac{a}{1-r}$.

So, $\frac{1}{2}+\frac{1}{4}+\ldots = \frac{\frac{1}{2}}{1-\frac{1}{2}} = \frac{\frac{1}{2}}{\frac{1}{2}} = 1$ Put values in (i)

Therefore, $\frac{1}{2}S = 1+1+1 \Rightarrow S = 6$

ii) $\mathbf{2+\frac{5}{3}+\frac{8}{9}+\frac{11}{27}+\ldots}$

Sol: Let $S = 2+\frac{5}{3}+\frac{8}{9}+\frac{11}{27}+\ldots$

Multiply both sides by $\frac{1}{3}$:

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$$\frac{1}{3}S=\frac{2}{3}+\frac{5}{9}+\frac{8}{27}+\frac{11}{81}+\ldots$$

Subtract the second equation from the first:

$$S-\frac{1}{3}S=\left(2+\frac{5}{3}+\frac{8}{9}+\frac{11}{27}+\ldots\right)-\left(\frac{2}{3}+\frac{5}{9}+\frac{8}{27}+\frac{11}{81}+\ldots\right)$$

$$\frac{2}{3}S=2+\left(\frac{5}{3}-\frac{2}{3}\right)+\left(\frac{8}{9}-\frac{5}{9}\right)+\left(\frac{11}{27}-\frac{8}{27}\right)+\ldots$$

$$\frac{2}{3}S=2+\frac{3}{3}+\frac{3}{9}+\frac{3}{27}+\ldots \Rightarrow \frac{2}{3}S=2+1+\frac{1}{3}+\frac{1}{9}+\ldots$$

The terms $\frac{1}{3}+\frac{1}{9}+\ldots$ form an infinite geometric series with the first term $a=\frac{1}{3}$ and the common ratio $r=\frac{1}{3}$. The sum of this series is $\frac{a}{1-r}$.

So, $\frac{1}{3}+\frac{1}{9}+\ldots=\frac{\frac{1}{3}}{1-\frac{1}{3}}=\frac{\frac{1}{3}}{\frac{2}{3}}=\frac{1}{2}$.

Therefore, $\frac{2}{3}S=2+1+\frac{1}{2}=3+\frac{1}{2}=\frac{7}{2}$

Multiplying by $\frac{3}{2}$, we get:

$$S=\frac{7}{2}\times\frac{3}{2}=\frac{21}{4}$$

So, the sum of the second series is $\frac{21}{4}$

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6. **Show that $2^{\frac{1}{2}}.4^{\frac{1}{4}}.8^{\frac{1}{8}}.16^{\frac{1}{16}}\ldots\infty=4$.**

Sol: Let the product be P:

$$P=2^{\frac{1}{2}}.4^{\frac{1}{4}}.8^{\frac{1}{8}}.16^{\frac{1}{16}}\ldots$$

Express the bases as powers of 2:

$$P=\left(2^1\right)^{\frac{1}{2}}.\left(2^2\right)^{\frac{1}{4}}.\left(2^3\right)^{\frac{1}{8}}.\left(2^4\right)^{\frac{1}{16}}\ldots$$

Simplify the terms using $\left(a^m\right)^n=a^{mn}$:

$$P=2^{1\left(\frac{1}{2}\right)}.2^{2\left(\frac{1}{4}\right)}.2^{3\left(\frac{1}{8}\right)}.2^{4\left(\frac{1}{16}\right)}\ldots$$

$$P=2^{\frac{1}{2}}.2^{\frac{2}{4}}.2^{\frac{3}{8}}.2^{\frac{4}{16}}\ldots$$

Combine the terms using $a^m.a^n=a^{m+n}$:

$$P=2^{\left(\frac{1}{2}+\frac{2}{4}+\frac{3}{8}+\frac{4}{16}+\ldots\right)}$$

Let the sum of the exponents be S:

$$S=\frac{1}{2}+\frac{2}{4}+\frac{3}{8}+\frac{4}{16}+\ldots=\sum_{n=1}^{\infty}\frac{n}{2^n}$$

To find the sum S, consider the geometric series:

$$G(x)=x+x^2+x^3+\ldots=\frac{x}{1-x},|x|<1$$

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Difference $G(x)$ with respect to x:

$$\frac{d}{dx}G(x)=1+2x+3x^2+\ldots=\frac{d}{dx}\left(\frac{x}{1-x}\right)=\frac{(1-x).1-x(-1)}{(1-x)^2}=\frac{1-\cancel{x}+\cancel{x}}{(1-x)^2}=\frac{1}{(1-x)^2}$$

Multiply by x:

$$x\left(1+2x+3x^2+\ldots\right)=x.\frac{1}{(1-x)^2}$$

$$x+2x^2+3x^3+\ldots=\frac{x}{(1-x)^2}$$

Substitute $x=\frac{1}{2}$:

$$S=\frac{1}{2}+2\left(\frac{1}{2}\right)^2+3\left(\frac{1}{2}\right)^3+\ldots=\frac{\frac{1}{2}}{\left(1-\frac{1}{2}\right)}$$

Substitute the value of S back into the expression for P:

$$P=2^S=2^2=4$$

Therefore, $2^{\frac{1}{2}}.4^{\frac{1}{4}}.8^{\frac{1}{8}}.16^{\frac{1}{16}}\ldots=4$

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7. Show that $\sqrt{4}.\sqrt[4]{16}.\sqrt[8]{64}.\sqrt[16]{256}\ldots\infty=16$

Sol: Let the product be P:

$$P=\sqrt{4}.\sqrt[4]{16}.\sqrt[8]{64},\sqrt[16]{256}\ldots$$

Rewrite the terms using factorial exponents and expressing the bases as powers of 4:

$$P=(4)^{\frac{1}{2}}.\left(4^2\right)^{\frac{1}{4}}.\left(4^3\right)^{\frac{1}{8}}.\left(4^4\right)^{\frac{1}{16}}\ldots$$

Simplify the terms using $\left(a^m\right)^n=a^{mn}$:

$$P=4^{1\left(\frac{1}{2}\right)}.4^{2\left(\frac{1}{4}\right)}.4^{3\left(\frac{1}{8}\right)}.4^{4\left(\frac{1}{16}\right)}\ldots$$

$$P=4^{\frac{1}{2}}.4^{\frac{2}{4}}.4^{\frac{3}{8}}.4^{\frac{4}{16}}\ldots$$

Combine the terms using $a^m.a^n=a^{m+n}$:

$$P=4^{\left(\frac{1}{2}+\frac{2}{4}+\frac{3}{8}+\frac{4}{16}+\ldots\right)}$$

Let the sum of the exponents be S:

$$S=\frac{1}{2}+\frac{2}{4}+\frac{3}{8}+\frac{4}{16}+\ldots=\sum_{n=1}^{\infty}\frac{n}{2^n}$$

To find the sum S, consider the geometric series:

$$G(x)=x+x^2+x^3+\ldots=\frac{x}{1-x},|x|<1$$

Difference $G(x)$ with respect to x:

$$\frac{d}{dx}G(x)=1+2x+3x^2+\ldots=\frac{d}{dx}\left(\frac{x}{1-x}\right)=\frac{(1-x)1-x(-1)}{(1-x)^2}=\frac{1-\cancel{x}+\cancel{x}}{(1-x)^2}=\frac{1}{(1-x)^2}$$

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Multiply by x:

$$x\left(1+2x+3x^2+\ldots\right)=x.\frac{1}{(1-x)^2}$$

$$x+2x^2+3x^3+\ldots=\frac{x}{(1-x)^2}$$

Substitute $x=\frac{1}{2}$:

$$S=\frac{1}{2}+2\left(\frac{1}{2}\right)^2+3\left(\frac{1}{2}\right)^3+\ldots=\frac{\frac{1}{2}}{\left(1-\frac{1}{2}\right)^2}=\frac{\frac{1}{2}}{\left(\frac{1}{2}\right)^2}=\frac{\frac{1}{2}}{\frac{1}{4}}=\frac{1}{2}\times\frac{4}{1}=2$$

Substitute the value of S back into the expression for P:

$P=4^S=4^2=16$

Therefore, $\sqrt{4}.\sqrt[4]{16}.\sqrt[8]{64}.\sqrt[16]{256}\ldots=16$

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8. Sum of n terms the series $2+4x+6x^2+8x^3+\ldots$ where $x\neq 1$

Sol: Let $S_n=2+4x+6x^2+8x^3+\ldots+2nx^{n-1}$

Multiply by x:

$xS_n=2x+4x^2+6x^3+8x^4+\ldots+2nx^n$

Subtract the second from the first:

$$S_n-xS_n=\left(2+4x+6x^2+8x^3+\ldots+2nx^{n-1}\right)-\left(2x+4x^2+6x^3+8x^4+\ldots+2nx^n\right)$$

$$S_n(1-x)=2+(4x-2x)+\left(6x^2-4x^2\right)+\ldots+\left(2nx^{n-1}-2(n-1)x^{n-1}\right)2nx^n$$

$$S_n(1-x)=2+2x+2x^2+\ldots+2x^{n-1}-2nx^n$$

$$S_n(1-x)=2\left(1+x+x^2+\ldots+x^{n-1}\right)-2nx^n$$

Use the geometric series sum formula:

$$1+x+x^2+\ldots+x^{n-1}=\frac{1-x^n}{1-x}$$

Substitute the sum:

$$S_n(1-x)=2\left(\frac{1-x^n}{1-x}\right)-2nx^n$$

$$S_n=\frac{2\left(1-x^n\right)-2nx^n(1-x)}{(1-x)^2}$$

$$S_n=\frac{2-2x^n-2nx^n+2nx^{n+1}}{(1-x)^2}$$

$$S_n=\frac{2-2(n+1)x^n+2nx^{n+1}}{(1-x)^2}$$

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9. Find the sum to n terms of the series: $\frac{2n+1}{2n-1}+3\left(\frac{2n+1}{2n-1}\right)^2+5\left(\frac{2n+1}{2n-1}\right)^3+\ldots$

Sol: Let $r=\frac{2n+1}{2n-1}$.

$S_n=r^1+3r^2+5r^3+\ldots$

Multiply by r:

$rS_n=r^2+3r^3+5r^4+\ldots$

Subtract the second from the first:

$S_n-rS_n=\left(r+3r^2+5r^3+\ldots\right)-\left(r^2+3r^3+5r^4+\ldots\right)$

$S_n(1-r)=r+2r^2+3r^3+2r^4+\ldots$

$S_n(1-r)=r+2\left(r^2+r^3+r^4+\ldots\right)$

For $|x|<1$, the geometric series

$x^2+x^3+x^4+\ldots=\frac{x^2}{1-x}$

$S(1-x)=x+2\left(\frac{x^2}{1-x}\right)$

$S(1-x)=\frac{x+x^2}{1-x}$

$S=\frac{x(1+x)}{(1-x)^2}$

Put $x=\frac{2n+1}{2n-1}$

$1+x=1+\frac{2n+1}{2n-1}=\frac{4n}{2n-1}$

$1-x=1-\frac{2n+1}{2n-1}=\frac{-2}{2n-1}$

$S=\frac{\left(\frac{2n+1}{2n-1}\right)\left(\frac{4n}{2n-1}\right)}{\left(\frac{-2}{2n-1}\right)^2}=n(2n+1)$

$S=\frac{\frac{(2n+1)(4n)}{(2n-1)^2}}{\frac{4}{(2n-1)^2}}=\frac{(2n+1)(4n)}{4}=n(2n+1)$

10. **Prove that $1+2\left(1+\frac{1}{n}\right)+3\left(1+\frac{1}{n}\right)^2+\ldots$ *to n terms* $=n^2$.**

Sol: Let $S_n=1+2\left(1+\frac{1}{n}\right)+3\left(1+\frac{1}{n}\right)^2+\ldots+n\left(1+\frac{1}{n}\right)^{n-1}$

Let $x=1+\frac{1}{n}$. Then $S_n=1+2x+3x^2+\ldots+nx^{n-1}$

Write the series: $S_n=1+2x+3x^2+\ldots+nx^{n-1}$

Multiply by x:

$xS_n=x+2x^2+3x^3+\ldots+(n-1)x^{n-1}+nx^n$

Subtract:

$S_n-xS_n=\left(1+2x+\ldots+nx^{n-1}\right)-\left(x+2x^2+\ldots+nx^n\right)$

$S_n-xS_n=1+x+x^2+\ldots+x^{n-1}-nx^n$

Geometric series sum:

$1+x+x^2+\ldots+x^{n-1}=\frac{1-x^{n-1}}{1-x}$

Substitute: $S_n(1-x)=\frac{1-x^n}{1-x}-nx^n$

Solve for S_n: $S_n=\frac{1-x^n}{(1-x)^2}-\frac{nx^n}{1-x}$

Substitute $x=1+\frac{1}{n}$ and $1-x=-\frac{1}{n}$

$$S_n=\frac{1-\left(1+\frac{1}{n}\right)^n}{\left(-\frac{1}{n}\right)^2}-\frac{n\left(1+\frac{1}{n}\right)^n}{\left(-\frac{1}{n}\right)}=\frac{1-\left(1+\frac{1}{n}\right)^n}{\left(-\frac{1}{n}\right)^2}+n^2\left(1+\frac{1}{n}\right)^n$$

$$S_n=n^2\left(1-\left(1+\frac{1}{n}\right)^n\right)+n^2\left(1+\frac{1}{n}\right)^n S_n=n^2-n^2\left(1+\frac{1}{n}\right)^n+n^2\left(1+\frac{1}{n}\right)^n$$

$S_n=n^2$

11. **Sum the series to n terms $2+5x+8x^2+11x^3+\ldots$ and deduce the sum to infinity if $|x|<1$.**

Sol: Sum to n terms:

Let $S_n=2+5x+8x^2+11x^3+\ldots+(3n-1)x^{n-1}$

Write xS_n:

$xS_n=2x+5x^2+8x^3+\ldots+(3n-4)x^{n-1}+(3n-1)x^n$

Subtract:

$S_n-xS_n=2+3x+3x^2+\ldots+3x^{n-1}-(3n-1)x^n$

$S_n-xS_n=2+3\left(x+x^2+\ldots+x^{n-1}\right)-(3n-1)x^n$

Geometric series sum:

$$x+x^2+\ldots+x^{n-1}=\frac{x\left(1-x^{n-1}\right)}{1-x}$$

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Substitute:

$$S_n(1-x) = 2 + 3\frac{x\left(1-x^{n-1}\right)}{1-x} - (3n-1)x^n$$

Solve for S_n :

$$S_n = \frac{2(1-x)+3x\left(1-x^{n-1}\right)-(3n-1)x^n(1-x)}{(1-x)^2}$$

$$S_n = \frac{2}{1-x} + \frac{3x\left(1-x^{n-1}\right)}{(1-x)^2} - \frac{(3n-1)x^n}{1-x}$$

Sum to infinity $(|x|<1)$:

As $n \to \infty, x_n \to 0$ & $x^{n+1} \to 0$

$$S_\infty = \lim_{n\to\infty} S_n = \frac{2}{1-x} + \frac{3x\left(1-x^\infty\right)}{(1-x)^2} - \frac{\left[3(\infty)-1\right]x^\infty}{1-x}$$

$$= \frac{2}{1-x} + \frac{3x(1-0)}{(1-x)^2} - \frac{\left[3(\infty)-1\right]0}{1-x}$$

$$= \frac{2}{1-x} + \frac{3x}{(1-x)^2} = \frac{2(1-x)+3x}{(1-x)^2}$$

$$= \frac{2-2x+3x}{(1-x)^2} = \frac{2+x}{(1-x)^2}$$

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